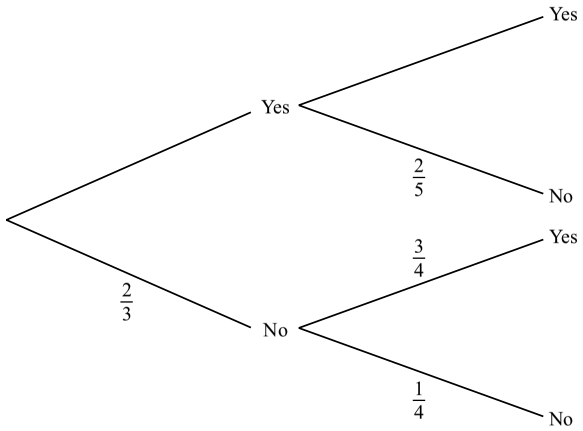
Question	Answer	Marks	Partial Marks
1	–2, –1, 0, 1	**2**	**B1** if one error/omission/extra
2	Correct triangle with arcs	**2**	**B1** correct no arcs
3	x^9y^4 final answer	**2**	**B1** for x^9y^k or x^ky^4 or correct answer seen then spoiled
4	–14, –5, 1	**2**	**B1** for 2 correct
5(a)	32.7 to 33.3	**1**	**FT** *their* printed diagram
5(b)	118 to 120	**1**	**FT** *their* printed diagram
5(c)	298 to 300	**2**	**B1** for 360 – (180 – *their* **(b)**) **FT** *their* **(b)** + 180
6(a)	5	**2**	**M1** for 10 × 12 × 'h' = 600
6(b)	172 or 172.4 to 172.5	**2**	**M1** for $1120 \times \frac{4}{3} \times \pi \times 0.45^3$
7(a)	20	**1**	
7(b)	 $\frac{2}{3}, \frac{2}{5}, \frac{3}{4}, \frac{1}{4}$ oe	**2**	**M1** for 2 or 3 correct
7(c)	$\frac{7}{10}$ oe	**3**	**M2** for $\frac{1}{3} \times \frac{3}{5}$ + *their* $\frac{2}{3}$ × *their* $\frac{3}{4}$ or **M1** for one product

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Question	Answer	Marks	Partial Marks
8(a)	$y = x$ ruled $x + y = 7$ ruled	3	**B1** for $y = x$ **B2** for $x + y = 7$ or **B1** for ruled line with gradient –1 or ruled line through (0, 7) or (7, 0) but not $x = 7$ or $y = 7$.
8(b)	Correct region below $y = x$, above $x + y = 7$ and above x-axis labelled R	1	
9(a)	$\frac{8}{3}$ oe	1	
9(b)	92	2	**M1** for correct calculation of an area
10	$(x + 3)(2 – 3y)$	2	**M1** for $2(x + 3) – 3y(x + 3)$ or $x(2y – 3) + 3(2y – 3)$
11(a)	6	2	**M1** for $10 + 13 – (20 – 3)$ oe or correctly completed Venn diagram
11(b)	7	1	**FT** *their* Venn diagram or 13 – *their* **(a)**
12	160.375	4	**M1** for mid-values for three of 125, 155, 162.5, 175 **M1** for Σfx where x is in correct interval including boundary **M1 dep** $\frac{\Sigma fx}{100}$ dep on second **M1**
13(a)	49.4 or 49.39 to 49.40	2	**M1** for $\tan = \frac{14}{12}$ oe
13(b)	22.4 or 22.36 to 22.37	4	**B1** for 62 + *their* **(a)** **M2** for $\sqrt{12^2 + 15^2 - 2 \times 12 \times 15 \cos(62 + \textit{their } \mathbf{(a)})}$ OR **B1** for 62 + *their* **(a)** **M1** for $12^2 + 15^2 – 2 \times 12 \times 15 \cos(62 + \textit{their } \mathbf{(a)})$ **A1** for 500.3 to 500.4
13(c)	13.2 or 13.24...	3	**M2** for $\sin 62 = \frac{\text{dist}}{15}$ or **M1** for recognition of shortest distance soi

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